

Total No. of Questions : 8]

SEAT No. :

**PD4591**

**[6404]-96**

[Total No. of Pages : 2

**B. E. (Computer Engineering)**  
**NATURAL LANGUAGE PROCESSING**  
**(2019 Pattern) (Semester - VIII) (410252(A)) (Elective - V)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, 7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

- Q1)** a) Explain the concept of log-linear models in NLP. How do these models work, and what are some common applications of this approach? [6]
- b) Explain the concept of doc2vec and its use in generating document embeddings. Discuss the differences between doc2vec and word2vec. [8]
- c) Explain Non-Negative Matrix Factorization (NMF) in the context of topic modeling [4]

OR

- Q2)** a) What is a Markov model, and how is it used to generate language sequences? Describe the process of training and using a Markov model for language generation. [6]
- b) Explain the concept of Latent Semantic Analysis (LSA) and how it is used for identifying relationships between words. [8]
- c) Write a short note on TF-IDF (Term Frequency-Inverse Document Frequency) representation. [4]

- Q3)** a) Explain the concept of reference resolution and coreference resolution in NLP. How do these techniques help in understanding the relationships between entities in a text? Provide examples. [8]
- b) Explain the concept of Cross-Lingual Information Retrieval (CLIR). Discuss the challenges of retrieving information across languages, and what techniques or models are commonly used to address these challenges? [8]
- c) What is Information Retrieval, and how is it used in NLP? [2]

OR

**P.T.O.**

- Q4)** a) Explain the process of Entity Extraction in Information Retrieval. Discuss the various techniques and algorithms used for Entity Extraction. [8]
- b) Describe the Vector Space Model (VSM) for information retrieval. How does VSM represent documents and queries, and how are similarities calculated? Discuss the strengths and weaknesses of VSM. [8]
- c) What is Named Entity Recognition (NER)? [2]

- Q5)** a) Explain how NLTK, spaCy, and Gensim handle tokenization and part-of-speech tagging. Discuss the advantages and disadvantages of each library's approach. [9]
- b) Explain the concept of WordNet as a lexical knowledge network. How can WordNet be used in natural language processing, and what are its limitations? [8]

OR

- Q6)** a) Discuss the role of Treebanks and Universal Dependency Treebanks in NLP research. How do they contribute to syntactic analysis and parsing in various languages? [9]
- b) Explain the Lesk Algorithm for word sense disambiguation. How does it work, and what are its advantages and limitations? [8]
- Q7)** a) Explain the key principles of rule-based machine translation. How do rule-based techniques differ from statistical approaches in machine translation? Provide an example of a rule-based translation. [9]
- b) Explain the fundamental components of a question-answering system. How do question answering systems utilize machine learning and NLP techniques to find accurate answers to questions? [8]

OR

- Q8)** a) Define text entailment and describe its importance in natural language understanding. How does text entailment differ from text similarity or paraphrasing? Provide an example of text entailment. [9]
- b) Explain Natural Language Generation with reference architecture. [8]

